

1 FM modulation

$$m(t) = 20 \sin c(200t), \quad c(t) = 100 \cos(2\pi f_c t)$$

$$1, 2) f(t) = f_c + f_{\Delta} m(t) = f_c + \frac{1}{2\pi} \frac{d\phi(t)}{dt}$$

$$\rightarrow f_{\Delta} m(t) = \frac{1}{2\pi} \frac{d\phi}{dt} \Rightarrow \phi(t) = 2\pi f_{\Delta} \int_0^t m(\tau) d\tau$$

$$\rightarrow x_c(t) = 100 \cos(2\pi f_c t + 2\pi f_{\Delta} \int_0^t 20 \sin c(200\tau) d\tau)$$

$$\beta \triangleq \frac{A_m f_{\Delta}}{f_m} = \frac{\text{max frequency deviation}}{m(t) \text{ bandwidth}}$$

$$\sin c \text{ at } \xrightarrow{F.T} \frac{1}{a} \text{ rect}\left(\frac{f}{a}\right) \Rightarrow F\{\sin c 200t\} = \frac{1}{200} \text{ rect}\left(\frac{f}{200}\right)$$

$$\rightarrow f_m = 100 \text{ Hz}, \quad A_m = \text{max}\{m(t)\} = 20$$

$$\rightarrow \beta = \frac{20 f_{\Delta}}{f_m} = \frac{20 f_{\Delta}}{100} = 4 \rightarrow f_{\Delta} = 20$$

$$\rightarrow x_c = 100 \cos(2\pi f_c t + 40\pi \int_0^t 20 \sin c 200\tau d\tau)$$

$$\text{max frequency dev.} = A_m f_{\Delta} = 400 \text{ Hz}$$

$$3) f_m = \omega = 100 \text{ Hz}$$

$$\rightarrow \beta_{FM} = \gamma(\beta+1)\omega = 2(4+1)100 = 1 \text{ KHz}$$

(carlson)

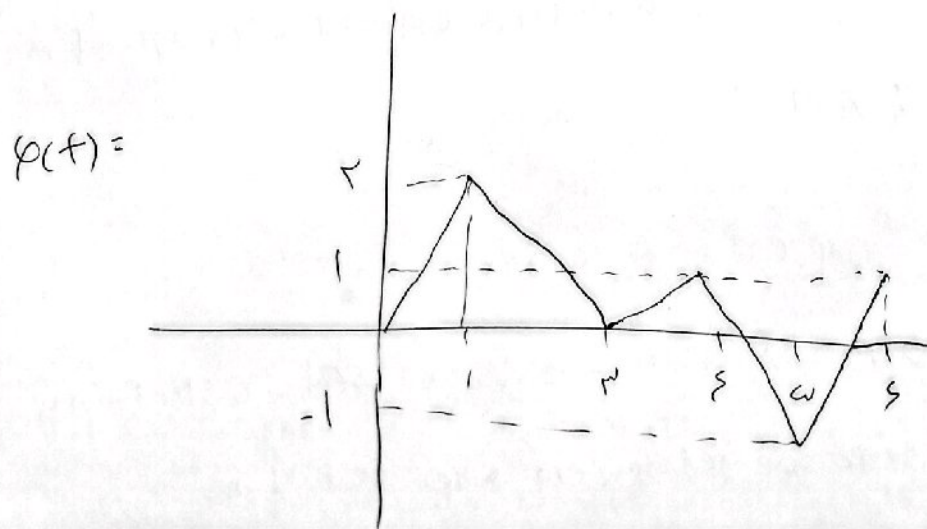
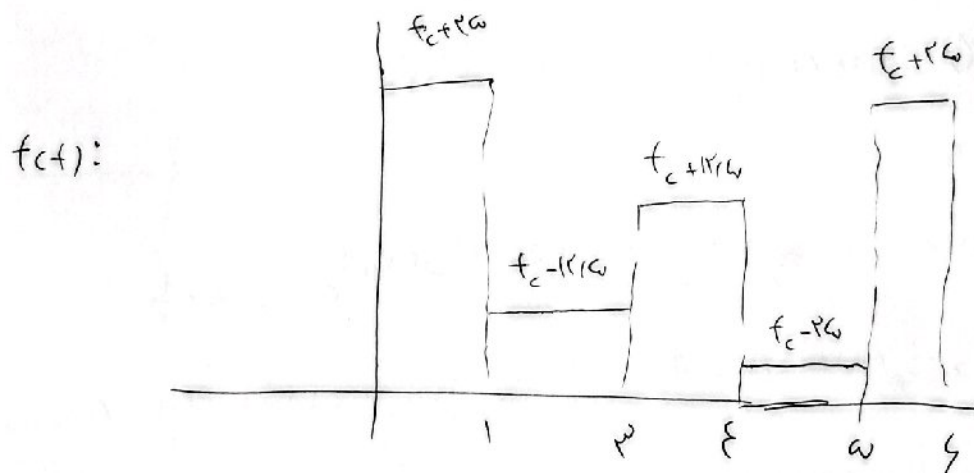
$$P_{ac} = \frac{A_c^2}{2} \sum_{n \in \mathbb{Z}} |J_n(\beta)|^2 = \frac{A_c^2}{2}, A_c = 100 \rightarrow P_{ac} = \frac{100^2}{2} = 5000$$

2 FM Modulator

$$f_{\Delta, \max} = A_m f_{\Delta}, A_m = \gamma, f_{\Delta} = \omega_{\max} \rightarrow f_{\Delta} = 1 \text{ KHz}$$

$$u_c(t) = A_c \cos\left(2\pi f_c t + 2\pi f_{\Delta} \int_0^t m(\tau) d\tau\right)$$

$$f(t) = f_c + f_{\Delta} m(t), \text{ Phase } = \varphi(t) = 2\pi f_{\Delta} \int_0^t m(\tau) d\tau$$



3 Frequency components of FM

1. if $c(t) = A_c \cos(2\pi f_c t)$, $m(t) = A_m \cos(2\pi f_m t)$

$$\rightarrow x_c(t) = A_c \cos\left(2\pi f_c t + 2\pi f_\Delta \int_0^t m(\tau) d\tau\right)$$

$$\rightarrow x_c(t) = A_c \cos\left(2\pi f_c t + \underbrace{\frac{A_m f_\Delta}{f_m}}_B \sin(2\pi f_m t)\right)$$

$$\rightarrow x_c(t) = A_c \sum_{n \in \mathbb{Z}} J_n(B) \cos(2\pi(f_c + f_m n)t)$$

$$\Rightarrow x_c(t) = \frac{A_c}{r} \sum_{n \in \mathbb{Z}} J_n(B) \left[\delta(t - t_c - n f_m) + \delta(t + t_c + n f_m) \right] \quad \textcircled{1}$$

$$\rightarrow P_{x_c} = \frac{A_c^2}{r} \sum_{n \in \mathbb{Z}} |J_n(B)|^2 = \frac{A_c^2}{r} \quad \star$$

Assigning values: $m(t) = \cos(20000\pi t)$, $c(t) = 100 \cos(2\pi f_c t)$

$$\rightarrow A_m = 1, A_c = 100, f_m = 10\text{K}, f_c = 100\text{MHz}$$

$$B = \frac{\text{Peak frequency deviation}}{\text{Bandwidth}} = \frac{20\text{K}}{10\text{K}} = 2$$

$$\star \rightarrow P_{x_c} = \frac{A_c^2}{r} = \frac{100^2}{r} = 2000 \rightarrow 10\% P_{x_c} = 200$$

$$P_{A_0} = \frac{A_c^2}{2} J_0^2(\beta) = \frac{A_c^2}{2} J_0^2(\gamma) = \frac{100^2}{2} (0.149)^2 \approx 100$$

$$P_{A_1} = \frac{100^2}{2} J_1^2(\gamma) \approx \frac{100^2}{2} (0.149)^2 \approx 144$$

$$P_{A_2} = \frac{100^2}{2} J_2^2(\gamma) \approx \frac{100^2}{2} (0.149)^2 \approx 4.8$$

$$P_{A_n} = \frac{100^2}{2} J_n^2(\gamma) \approx \frac{100^2}{2} (0.149)^2 \approx 10$$

$$, P_{A_n} \text{ if } n > 2 < 10 \Rightarrow \text{only } P_{A_1}, P_{A_2}, \text{ i.e. } P_{A_c} = 100$$

$$\textcircled{1} \rightarrow A_1: \text{ Amplitude} = \frac{A_c}{2} J_1(\gamma) \approx 1.118 \text{ V}$$

$$f = f_c + 1 f_m, f_c - 1 f_m \rightarrow f = 100.01, 99.99 \text{ MHz}$$

$$A_2: \text{ Amplitude} = \frac{A_c}{2} J_2(\gamma) \approx 1 \text{ V}$$

$$f = f_c + 2 f_m, f_c - 2 f_m \rightarrow f = 100.2, 99.8 \text{ MHz}$$

$$2. B_{FM} = 2(1+\beta) f_m = 2(1+1) 10 \text{ K} = 40 \text{ KHz}$$

4) FM signal power

$$1. x_c(t) = A_c \cos(2\pi f_c t + \phi(t)), P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^T |x_c(t)|^2 dt$$

$$\rightarrow P = \lim_{T \rightarrow \infty} \int_{-T}^T A_c^2 \cos^2(2\pi f_c t + \phi(t)) dt, \cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\rightarrow P = A_c^2 \lim_{T \rightarrow \infty} \int_{-T}^T \frac{1}{2} dt + \frac{A_c^2}{2} \lim_{T \rightarrow \infty} \int_{-T}^T \cos(4\pi f_c t + 2\phi) dt$$

$$= \frac{A_c}{r} + I$$

$$2. \quad \cos \theta = \frac{1}{r} \int e^{j\theta} + e^{-j\theta}$$

$$\rightarrow \frac{I}{\infty} = \int_{-\infty}^{\infty} \cos(\xi \pi t_c + \varphi) dt = \frac{1}{r} \left[\int_{-\infty}^{\infty} e^{j(\xi \pi t_c + \varphi)} dt + \int_{-\infty}^{\infty} e^{-j(\xi \pi t_c + \varphi)} dt \right]$$

$$= \frac{1}{r} \left[F \left\{ e^{j\varphi(t)} \right\}_{t=\xi \pi t_c} + F \left\{ e^{-j\varphi(t)} \right\}_{t=-\xi \pi t_c} \right]$$

3.

$$\text{Taylor series at } t=0: \varphi(t) = \varphi(0) + \varphi'(0)t + \frac{\varphi''(0)}{2}t^2 + \dots$$

$$\rightarrow \cos(\xi \pi t_c + \varphi(t)) = \cos\left(\xi \pi t_c + \varphi(0) + \varphi'(0)t + \frac{\varphi''(0)}{2}t^2 + \dots\right)$$

$$\text{trigonometric expansion} \quad = \cos(\xi \pi t_c + \varphi(0)) - \sin(\xi \pi t_c + \varphi(0)) \varphi'(0) + \dots$$

$$\rightarrow F\{\varphi'(t)\} \text{ contributes to the frequencies } \pm \xi \pi t_c$$

$$4. \quad \frac{I}{\infty} = \int_{-\infty}^{\infty} \cos(\xi \pi t_c + \varphi(0)) - \sin(\xi \pi t_c + \varphi(0)) \cdot \varphi'(0) + \dots$$

$$F\{\varphi(t)\} = \int_{-\infty}^{\infty} \varphi(t) e^{-j\omega t} dt \rightarrow F\{\varphi'(t)\} = \int_{-\infty}^{\infty} (j\omega) F\{\varphi(t)\}$$

$$\rightarrow t_c \gg \omega \Rightarrow \int_{-\infty}^{\infty} \approx 0$$

5. in part 1, we had $P = \frac{A_c^2}{2} + I$

$$\text{if } t_c \gg \omega \rightarrow I_\infty \approx 0 \rightarrow P = \frac{A_c^2}{2}$$

5 Envelope Detector for FM signal

$$s(t) = A_c \cos(\omega_c t + \omega_c K_f \int_0^t m(\tau) d\tau)$$

$$\text{HP filter F.T.} = \frac{j\omega RC}{1 + j\omega RC} \quad |\omega RC| \ll 1$$

$$\rightarrow H(j\omega) \approx j\omega RC \quad (\text{a differentiator})$$

$$\rightarrow y(t) = RC \frac{ds}{dt} \quad (\text{HP output})$$

$$\rightarrow y(t) = RC A_c \sin(\omega_c t + \omega_c K_f \int_0^t m(\tau) d\tau) (\omega_c t + \omega_c K_f \int_0^t m(\tau) d\tau)$$

$$K_f |m(t)| < f_c \Rightarrow y(t) = RC A_c \omega_c (\omega_c t + K_f \int_0^t m(\tau) d\tau)$$

6 Frequency spectrum of an FM signal

$$m(t) = A \cos \pi f_1 t + B \cos \pi f_2 t$$

$$\rightarrow x_c(t) = A_c \cos(\pi f_c t + \beta_1 \sin \pi f_1 t + \beta_2 \sin \pi f_2 t)$$

$$\rightarrow x_c(t) = A_c \operatorname{Re} \left\{ \exp(j\pi f_c t + j\beta_1 \sin \pi f_1 t + j\beta_2 \sin \pi f_2 t) \right\}$$

from F.S we know that: $e^{j\beta \sin \pi f_m t} = \sum_{n \in \mathbb{Z}} J_n(\beta) e^{j\pi n f_m t}$

$$\rightarrow x_c = A_c \operatorname{Re} \left\{ e^{j\pi f_c t} \sum_{n \in \mathbb{Z}} J_n(\beta_1) e^{j\pi n f_1 t} \sum_{m \in \mathbb{Z}} J_m(\beta_2) e^{j\pi m f_2 t} \right\}$$

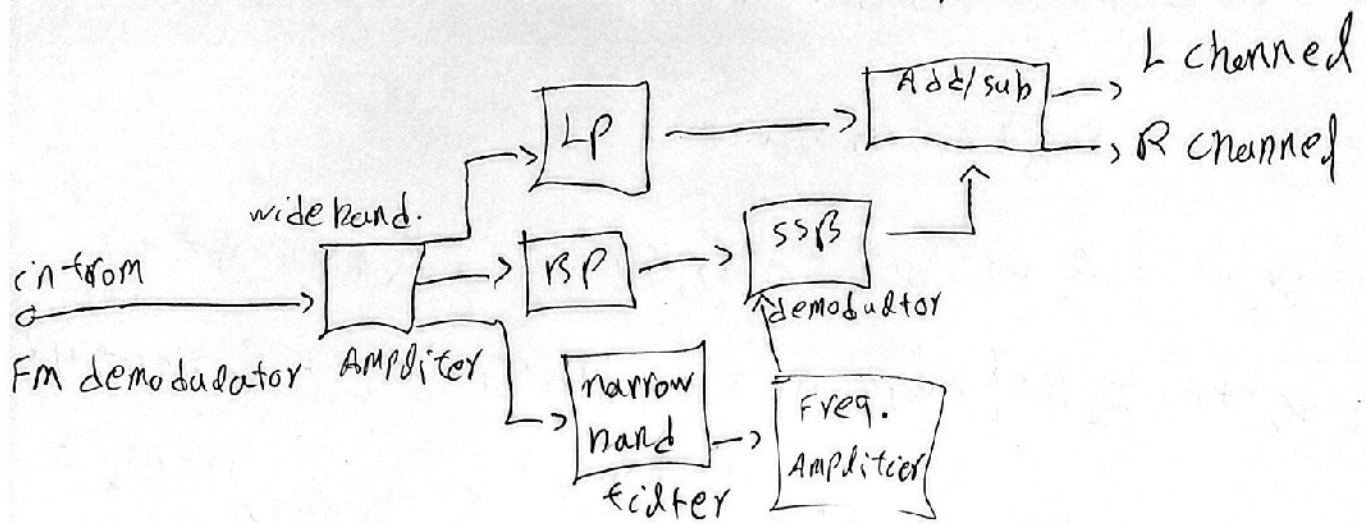
$$\xrightarrow{\text{F.T}} X_c(f) = \frac{A_c}{r} \sum_{n \in \mathbb{Z}} \sum_{m \in \mathbb{Z}} J_n(\beta_1) J_m(\beta_2) \left[\delta(f - f_c - n f_1 - m f_2) + \delta(f + f_c + n f_1 + m f_2) \right]$$

$\rightarrow$ deltas at $f = \pm(f_c + n f_1 + m f_2)$ with Amplitude of $J_n(\beta_1) J_m(\beta_2)$

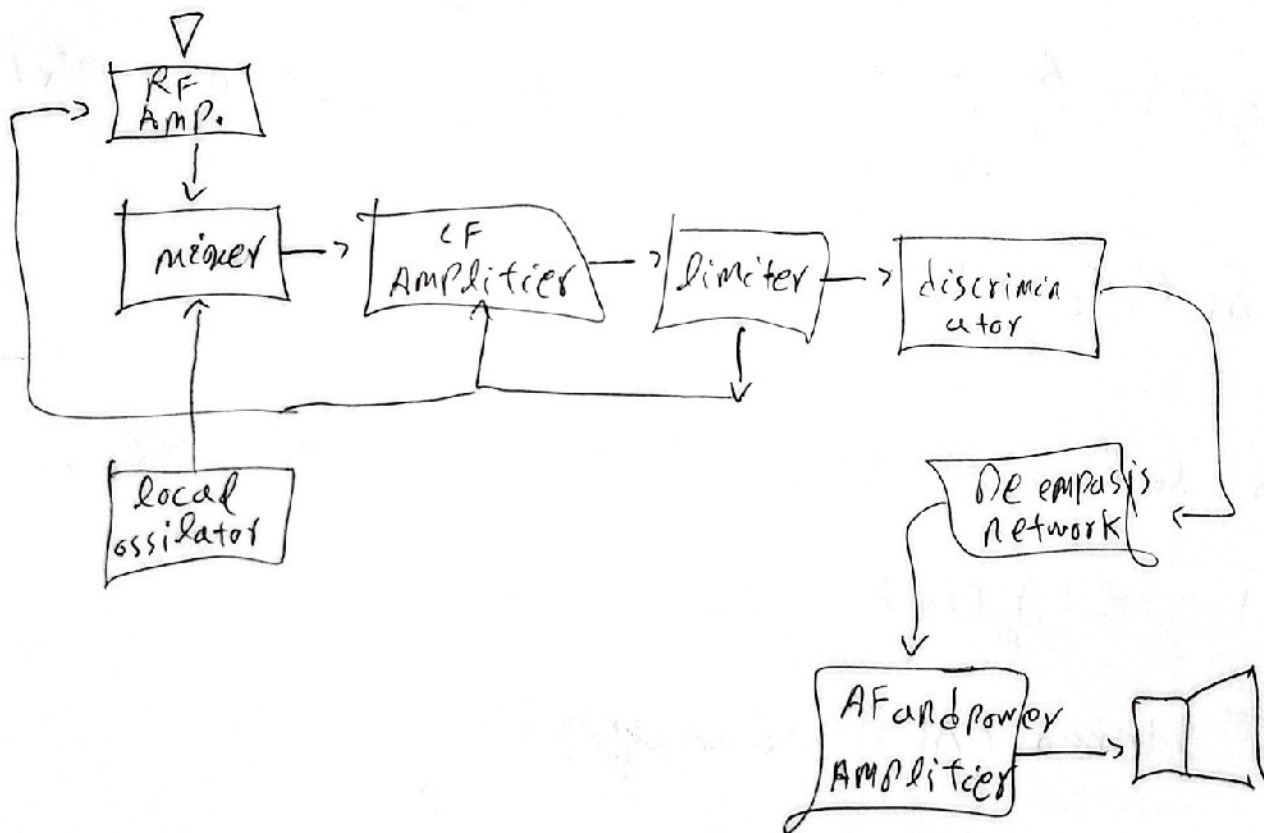
7 Stereo FM Broadcasting

stereo FM Broadcasting transmits two separate audio using a single FM carrier. the FM transmitter encodes the stereo signal and FM receiver decodes it

Providing Stereo Sound on the output



FM multiplex demodulation



3 superheterodyne Receiver

$$f_{LF} = |f_{LO} - f_c|, \quad f_c' = f_{LO} \pm f_{IF}$$

$\uparrow$
image frequency

$$f_{LO} + f_{IF} > 108 \text{ MHz}, \quad f_{IF} < f_{LO}$$

$$\Rightarrow f_{LO} = 108 \text{ MHz}$$

$$f_c' = f_{LO} + f_{IF} > 108 \text{ MHz} + f_{IF}$$

$$\rightarrow f_{IF} \geq 18 \text{ MHz}$$

corresponding range: $f_{LO} \pm f_{IF}$

$$\rightarrow 108 \text{ MHz} \leq f_{LO} < 126 \text{ MHz}$$